Kaduna.lk - Unified Smart Platform for Roadside Assistance and Automotive Services in Sri Lanka

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Project Proposal Report

Janukshan Sivakumar – IT22635266

B.Sc. (Hons) Degree in Information Technology Specialized in Software Engineering

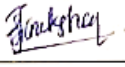
Department of Software Engineering

Sri Lanka Institute of Information Technology Sri Lanka

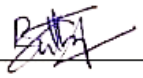
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Janukshan Sivakumar	IT22635266	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Ms. Karthiga Rajendran	16-03-2026	
Ms. Kaushalya Rajapakshe	16/03/2026	

ABSTRACT

Vehicle breakdowns and roadside emergencies are frequent on Sri Lankan roads, yet the country lacks a dedicated digital platform for managing roadside assistance dispatch, forcing drivers to rely on manual phone calls to insurance hotlines, personal contacts, or nearby garages, which often results in long delays, mismatched service providers, and worsened traffic congestion. Unlike the USA and parts of Europe, where platforms such as Honk and Urgently streamline roadside dispatch, Sri Lanka has no equivalent system, making this research a pioneering effort to introduce an intelligent dispatch solution. A critical gap in existing systems, both local and international, is their reliance on simple nearest-provider assignment, assuming the service type is known at request time, whereas in reality, a reported "vehicle won't start" could stem from a dead battery, faulty starter motor, or fuel pump failure, each requiring different expertise. Dispatching the wrong provider wastes time and resources while exacerbating congestion. To address this, the proposed Uncertainty-Aware Dispatch Optimization (UADO) framework treats dispatch as a stochastic optimization problem under service-type uncertainty. It integrates a Diagnostic Triage System that combines structured questionnaires with real-time OBD-II vehicle telemetry to estimate probabilities of different service needs, an Expected-Cost Minimization algorithm that selects providers by weighing mismatch risks and traffic impact, and an Adaptive Bayesian Learning mechanism that continuously improves diagnostic accuracy based on past outcomes. Developed using Node.js, TypeScript, PostgreSQL, Python, and Google Maps API, the system is evaluated through simulations of 1,000 synthetic incidents in the Colombo metropolitan area, with expected results showing a 35–45% reduction in resource mismatch rates and a 20–30% improvement in resolution times. As Sri Lanka's first intelligent roadside dispatch system, this research promises to reduce driver wait times, alleviate traffic congestion caused by breakdowns, and lay the foundation for a more organized, efficient, and reliable roadside service industry in the country.

Keywords: Dispatch Optimization, Roadside Assistance, Stochastic Optimization, Diagnostic Triage, OBD-II Telemetry

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Table of Contents

DECLARATION 1

ABSTRACT 2

ACKNOWLEDGEMENT 3

LIST OF TABLES 5

LIST OF FIGURES 5

LIST OF ABBREVIATIONS 6

1. INTRODUCTION 7

1.1 Background and Context 7

1.2 Literature Review 8

1.3 Research Gap 11

2. OBJECTIVE 12

2.1 Main Objective 12

2.2 Specific Objectives 12

3. METHODOLOGY 13

4. HIGH-LEVEL SYSTEM ARCHITECTURE 17

5. USER REQUIREMENTS 18

6. COMMERCIALIZATION PLAN 20

7. BUDGET AND JUSTIFICATION 21

8. WORK BREAKDOWN STRUCTURE (WBS) 22

9. GANTT CHART 23

10. REFERENCES LIST 24

11. APPENDICES 25

LIST OF TABLES

Table 1- List of Abbreviations 6
Table 2- Summary of Related Work 10
Table 3 - Mapping of Objectives to Deliverables 12
Table 4 - Tools and Technologies 15
Table 5 - Evaluation Metrics and Targets 16
Table 6 - Component Boundary 17
Table 7 - Stakeholder Identification 18
Table 8 - Functional Requirements 19
Table 9 - Non-Functional Requirements 19
Table 10 - Budget Breakdown 21
Table 11 – Work Breakdown 22
Table 12 - Diagnostic Triage Sample Question Set 26
Table 13 - Provider Type Capability Matrix 27

LIST OF FIGURES

Figure 1- Dispatch Service Internal Architecture 17
Figure 2 - Dispatch Service Data Flow 18
Figure 3 - Gantt Chart 23
Figure 4 - UADO Framework Overview 25
Figure 5 - Expected Cost Minimization Process 28

LIST OF ABBREVIATIONS

Abbreviation	Full Form
UADO	Uncertainty-Aware Dispatch Optimization
ECM	Expected-Cost Minimization
OBD-II	On-Board Diagnostics (Second Generation)
DTC	Diagnostic Trouble Code
ETA	Estimated Time of Arrival
API	Application Programming Interface
GPS	Global Positioning System
MCDM	Multi-Criteria Decision Making
MILP	Mixed-Integer Linear Programming
ML	Machine Learning
IoT	Internet of Things
SLA	Service Level Agreement
PDPA	Personal Data Protection Act
VRP	Vehicle Routing Problem
RUL	Remaining Useful Life
SDG	Sustainable Development Goal

Table 1- List of Abbreviations

1. INTRODUCTION

1.1 Background and Context

Vehicle breakdowns and roadside emergencies are everyday occurrences on Sri Lankan roads, yet the country still lacks a dedicated digital system to manage roadside assistance dispatch. At present, when a vehicle breaks down or is involved in an accident, drivers must rely on manual phone calls to insurance hotlines, personal contacts, or nearby garages to arrange help. This fragmented, phone-based approach results in average wait times of 45 to 90 minutes in urban areas and significantly longer in rural regions. The consequences extend beyond the stranded driver. Service providers frequently waste trips due to mismatched equipment, as the wrong type of provider is often sent without a proper diagnostic assessment. More critically, delayed incident clearance on busy roads causes cascading traffic congestion that affects thousands of other road users. Research by demonstrates that reducing incident duration by just 50% can cut total traffic delays by up to 75%, highlighting the significant public cost of inefficient roadside response [17].

While countries such as the United States have digital roadside dispatch platforms like Honk and Urgently, these systems rely on deterministic matching based on user-reported incident categories. They assume the required service type is known at the time of the request, which is often not the case. A driver reporting "my car won't start" could need a simple battery jump-start, a tow due to starter motor failure, or fuel delivery, each requiring a completely different provider. These platforms also lack integration with real-time vehicle telemetry data and do not account for the broader social cost of traffic impact when making dispatch decisions. Notably, Sri Lanka has no equivalent system, whether commercial or research-based, making this a first-time effort to introduce intelligent dispatch optimization for roadside assistance in the country.

To address these gaps, this research proposes the Uncertainty-Aware Dispatch Optimization (UADO) framework, built on a Predict-Then-Optimize paradigm. The framework consists of three core parts: a Diagnostic Triage System that combines structured questionnaires with real-time OBD-II vehicle data to estimate the probability of different service needs, an Expected-Cost Minimization (ECM) algorithm that selects the optimal provider by accounting for mismatch risks and traffic externality costs, and an Adaptive Bayesian Learning mechanism that continuously refines diagnostic accuracy based on past resolution outcomes. Studies in related domains show that multi-criteria dispatching can achieve up to 19% reduction in operational costs [1], supporting the viability of this approach. The system is evaluated through simulation on 1,000 synthetic incidents across the Colombo metropolitan area over a 14-day period.

This research aligns with SDG 9 (Industry, Innovation, and Infrastructure) and SDG 11 (Sustainable Cities and Communities) by digitizing transport services and reducing urban congestion caused by roadside incidents. It is conducted under the Software Systems and Technologies (SST) research cluster as part of the Kaduna.lk project, which is vertically integrated with Geo-Intelligence, Predictive Maintenance, and Compliance components to form a unified smart platform for roadside assistance in Sri Lanka.

1.2 Literature Review

The Kaduna.lk platform represents a holistic, smart ecosystem designed to modernize roadside assistance and automotive services in Sri Lanka through a multi-component integration. The system is built upon four interconnected research pillars: Dispatch Optimization and Smart Recovery, which handles resource allocation under uncertainty; Geo-Intelligence and Traffic Impact Analysis, which provides spatial data on incident-induced congestion; Predictive Maintenance, leveraging AI-driven vehicle health monitoring to preempt failures; and Compliance and Stakeholder Engagement, ensuring the platform adheres to the Personal Data Protection Act (PDPA) while fostering ecosystem integration.

The development of the Uncertainty-Aware Dispatch Optimization (UADO) framework is grounded in a multi-disciplinary review of current commercial roadside assistance platforms, academic dispatch models, vehicle telematics, and decision-making theories under uncertainty.

1.2.1 Roadside Assistance Dispatch Systems

Several commercial platforms provide digital roadside assistance dispatch. Honk (USA) is a B2B platform connecting insurance companies with roadside providers using location-based matching. Urgently (USA) offers API-based dispatch for enterprise clients with real-time ETA tracking. Both systems rely on user-reported incident categories and use deterministic nearest-match dispatch without modeling service-type uncertainty [18][19].

The EmerGuard project from Mumbai University (2024) developed a mobile app with GPS and a hardware emergency button to connect stranded drivers with nearby providers. While it demonstrated the feasibility of location-based emergency request systems, it did not address the resource mismatch problem or incorporate diagnostic triage [20].

On the academic side a multi-objective optimization system for RACC (Royal Automobile Club of Catalonia, Spain) that automated 100% of dispatch assignments and achieved 19% cost reduction. Their approach considered skill matching, priorities, and cost using heuristics and linear programming. However, their model assumed the service type was known at dispatch time—the optimization was over which provider to send, not over uncertainty about what service was needed [1].

1.2.2 Emergency Vehicle Dispatch and Ambulance Allocation

The emergency medical services (EMS) literature provides relevant methodological foundations. A dynamic penalty-based dispatching approach (Frontiers, 2025) quantified supply-demand levels within each ambulance's service area before dispatch, achieving 90% of high-priority calls with response times under 6 minutes versus 75% with traditional greedy methods [4]. This externality-aware approach informs the traffic impact integration in this research.

Research combining ML with combinatorial optimization for ambulance dispatching achieved up to 30% reduction in mean response times compared to industry benchmarks. A study on

multi-criteria assignment for EMS with non-homogeneous specialty (Scientific Reports, 2023) addressed the problem of matching ambulances with different capability levels to patients with different needs a heterogeneous fleet matching problem analogous to roadside assistance [7]. Vanderbilt University's (2024) stochastic dispatch framework using survival modeling for incident prediction represents the closest methodological precedent to the UADO framework proposed here [2].

1.2.3 Vehicle Fault Diagnosis and OBD-II Integration

Modern vehicle telematics enable remote fault detection that can inform dispatch decisions. A hybrid ML and RAG approach (Springer, 2025) combined generative AI with OBD-II diagnostics to produce comprehensive diagnostic reports with 85% readability scores [12]. STREAM-VAE developed a dual-path variational autoencoder for real-time anomaly detection in automotive telemetry [13]. Geotab's (2025) commercial system enriches raw OBD fault codes with severity ratings and recommended actions [16]. These advances demonstrate the feasibility of using vehicle data to reduce diagnostic uncertainty before dispatch.

An AI-based predictive maintenance study using OBD-II (IRJMETS, 2024) demonstrated that AI systems can identify incipient faults before Diagnostic Trouble Codes are triggered, enabling early warnings rather than post-failure alerts [15]. This proactive detection capability supports integration with the Predictive Maintenance component.

1.2.4 Predict-Then-Optimize Frameworks

The operations research literature on decision-making under uncertainty provides theoretical foundations for this work. A comprehensive review of predict-and-optimize techniques validated the paradigm where machine learning predictions feed directly into optimization problems [9]. The ASTER framework coupled early prediction accuracy with actionable decision-making through resource-aware optimization [3]. A scenario-based approach for inventory routing used quantile forecasts with stochastic programming, aligning with the diagnostic triage approach of producing probability distributions rather than point predictions [10].

1.2.5 Traffic Impact of Incidents

VISSIM traffic simulation to show that reducing incident duration by 50% can cut total traffic delays by up to 75% [17]. This non-linear relationship justifies incorporating traffic externality costs in dispatch optimization prioritizing incidents on busy roads were delays cascade into widespread congestion. Empirical analysis consistently shows that unplanned breakdowns significantly increase delays and reduce road capacity, with effects amplified on roads with limited alternative routes a common situation in Sri Lanka.

Domain	Source	Approach	Key Finding	Limitation
Roadside Dispatch	Buil et al. (2021) [1]	Multi-objective optimization, heuristics	100% automated, 19% cost reduction	Assumes known service type
Roadside App	EmerGuard (2024) [20]	GPS mobile app + hardware button	Validates location-based assistance	Manual dispatch; no optimization
EMS Dispatch	Vanderbilt (2024) [2]	Stochastic dispatch, survival modeling	Predict-then-optimize for ambulances	Applied to EMS, not roadside
EMS Allocation	Scientific Reports (2023) [7]	Multi-criteria, heterogeneous fleet	Capability-aware matching	Medical context only
Vehicle Diagnosis	Hybrid ML+RAG (2025) [12]	OBD-II + generative AI	85% diagnostic readability	Not integrated with dispatch
Predict-Optimize	ASTER (2025) [3]	Early decision + resource optimization	Resource-aware interventions	General framework, not roadside
Traffic Impact	Koliou et al. (2024) [17]	VISSIM simulation, V2V detection	50% faster clearance = 75% less delay	UK motorways; no scoring model
Commercial	Honk, Urgently [18][19]	Nearest-match dispatch	Industry-standard platforms	Deterministic; no uncertainty modeling

Table 2- Summary of Related Work

1.3 Research Gap

Despite extensive work on dispatch optimization in ride-hailing and emergency medical services, significant gaps remain when applying these methods to roadside assistance. The literature review reveals five critical gaps:

Gap 1: No Dispatch Optimization Under Service-Type Uncertainty

Existing roadside assistance systems (Honk, Urgently, EmerGuard) and academic studies all assume the service type is known with certainty at dispatch time. They optimize which provider to send, but not what to do when there is uncertainty about what service is needed. In practice, a significant portion of roadside incidents involve diagnostic ambiguity. A driver who reports "my car won't start" could have a dead battery, a failed starter motor, a fuel pump failure, or an empty fuel tank each requiring a different type of provider. Dispatching based on the most likely diagnosis without accounting for uncertainty leads to high mismatch rates. This research formulates dispatch as a stochastic optimization problem where the input is a probability distribution over service types, and the objective is to minimize expected total cost including mismatch penalties.

Gap 2: No Integration of Remote Diagnostic Data in Dispatch Decisions

No existing system integrates real-time vehicle telemetry (OBD-II data) with dispatch decisions. Commercial telematics platforms like Geotab use OBD data for fleet maintenance scheduling, but this information is not used to inform emergency dispatch. If a vehicle's OBD system reports a P0300 code (random misfire) with low battery voltage, the dispatch system could be much more confident about the diagnosis than from user-reported symptoms alone. This research integrates OBD-II telemetry from the Predictive Maintenance component as an input to the diagnostic triage system, combining user-reported symptoms with objective vehicle health data.

Gap 3: No Externality-Aware Dispatch for Roadside Assistance

Ride-hailing dispatch optimizes for individual user wait time. Roadside assistance has a public externality: a stranded vehicle on a major road causes congestion affecting thousands of other road users. No existing roadside system incorporates this externality in dispatch decisions. If two incidents occur simultaneously one on a quiet residential street, one on the Southern Expressway during rush hour the system should prioritize the expressway incident even if the residential one was reported first. This research integrates traffic impact scores from the Geo-Intelligence component as externality weights in the dispatch objective function.

Gap 4: No Adaptive Learning from Resolution Outcomes

Existing dispatch systems do not learn from the gap between predicted and actual service requirements. If the system consistently over-predicts towing needs for certain symptom combinations, it has no mechanism to self-correct. Diagnostic accuracy can improve significantly with feedback. This research implements Bayesian updating of diagnostic probabilities based on resolution outcomes, creating a self-improving system where each resolved incident refines future predictions.

2. OBJECTIVE

2.1 Main Objective

To design, implement, and evaluate an Uncertainty-Aware Dispatch Optimization (UADO) framework that formulates roadside assistance dispatch as a stochastic optimization problem, integrating diagnostic triage with OBD-II vehicle telemetry, externality-aware cost functions, and adaptive Bayesian learning to minimize expected total resolution cost including resource mismatch penalties and traffic impact costs.

2.2 Specific Objectives

- **SO1: Design and Validate a Multi-Source Diagnostic Triage System –** Develop a diagnostic triage system that combines structured user-reported symptoms with real-time OBD-II vehicle telemetry data to produce probability distributions over possible service types, reducing dispatch uncertainty. Target: highest-probability diagnosis matches actual outcome in ≥70% of test cases.
- **SO2: Formulate and Implement the Stochastic Dispatch Model –** Develop the Expected-Cost Minimization (ECM) algorithm that optimizes provider selection by minimizing expected total resolution cost under service-type uncertainty, including mismatch penalties and traffic externality costs. Target: dispatch decision in <500ms.
- **SO3: Develop Adaptive Bayesian Learning Mechanism –** Implement a feedback loop that updates diagnostic probabilities based on resolution outcomes, enabling the system to improve accuracy over time. Target: measurable improvement after ~150 resolved incidents.
- **SO4: Integrate with Predictive Maintenance and Geo-Intelligence Components –** Establish integration points with the Predictive Maintenance component (for OBD-II data) and the Geo-Intelligence component (for traffic impact scores) through well-defined API contracts.
- **SO5: Evaluate Framework Through Simulation –** Design and execute a comprehensive simulation study comparing UADO against baseline dispatch strategies on 1,000 synthetic incidents. Target: ≥30% reduction in mismatch rate, ≥15% reduction in mean resolution time ($p < 0.05$).

Objective	Key Deliverables	Evaluation Method
Diagnostic Triage	Decision tree specification, OBD mapping table, Validation report	Accuracy testing against labeled dataset
Stochastic Dispatch	Mathematical formulation, ECM implementation, Sensitivity analysis	Simulation comparison with baselines
Adaptive Learning	Bayesian algorithm, Learning curve analysis	Accuracy improvement over time
Component Integration	API specifications, Integration tests	End-to-end data flow verification
Simulation Evaluation	Dataset, Simulation environment, Results report	Statistical significance testing

Table 3 - Mapping of Objectives to Deliverables

3. METHODOLOGY

This research follows a Design Science Research (DSR) approach combined with experimental simulation methodology. The focus is on designing and evaluating a software artifact - the UADO dispatch framework that provides optimal resource allocation decisions under uncertainty. The methodology is structured into six phases, each producing verifiable outputs.

3.1 Research Questions

RQ1: Does a structured diagnostic triage combining user symptoms with OBD-II data significantly reduce service-type uncertainty compared to user-reported categorization alone?

RQ2: Does Expected-Cost Minimization reduce total resolution times and mismatch rates compared to deterministic dispatch strategies?

RQ3: How does incorporating traffic externality costs affect resource allocation and network-level outcomes?

RQ4: Does Bayesian updating of diagnostic probabilities measurably improve dispatch accuracy over time?

3.2 Hypotheses

H1: Multi-source diagnostic triage (questionnaire + OBD-II) produces probability distributions with significantly lower entropy than user-reported categories alone.

H2: ECM dispatch achieves at least 30% reduction in mismatch rate and 15% reduction in mean resolution time compared to type-matched nearest dispatch ($p < 0.05$).

H3: Incorporating traffic externality costs (lambda > 0) reduces total network vehicle-hours lost by at least 15% compared to individual-cost-only optimization.

H4: Bayesian learning achieves measurable accuracy improvement (at least 10%) after approximately 150 resolved incidents.

3.3 Research Approach

3.3.1 Diagnostic Triage System Development

The diagnostic triage system addresses a fundamental question: what service does this incident need? Rather than accepting the user's self-reported category, the system gathers structured information to produce a probability distribution over possible service types.

- **Structured Diagnostic Questionnaire:** A decision tree-based questionnaire designed from expert knowledge to efficiently narrow down the diagnosis.
- **OBD-II Data Integration:** For vehicles with connected OBD-II devices, the system retrieves real-time diagnostic data.
- **Probability Distribution Output:** The diagnostic engine combines questionnaire responses and OBD-II data to produce a probability distribution over nine service type categories.

3.3.2 Stochastic Dispatch Optimization

Given an incident with a service type probability distribution from the diagnostic triage, the system selects the optimal provider to dispatch.

Expected-Cost Minimization (ECM) Algorithm:
The algorithm evaluates each provider by computing expected costs across possible service outcomes, applies penalties for mismatches when a provider cannot deliver the required service, incorporates traffic externality costs from the Geo-Intelligence module, adjusts decisions using trust scores based on provider history, and achieves real-time responsiveness by running in O(n\times k) time, ensuring dispatch decisions are completed in under 500 milliseconds.

Baseline Strategies for Comparison:

1. Greedy-Nearest: Select the closest available provider regardless of type
2. Type-Matched Nearest: Select the closest provider that can handle the most likely service type
3. Deterministic Multi-Criteria: Weighted scoring treating service type as deterministic

3.3.3 Adaptive Bayesian Learning

Every resolved incident in the system generates a feedback record that captures the predicted service type distribution, the actual service type delivered, the mismatch status, resolution time, and user rating, creating a rich dataset for continuous improvement. The learning mechanism maintains prior probabilities for each symptom-to-diagnosis mapping and updates these priors using Bayes' theorem after each resolved incident, ensuring that diagnostic accuracy improves with experience. To balance adaptability with stability, the learning rate begins at 0.1 and gradually decays to 0.01, preventing overcorrection while still allowing refinement. Additionally, the system tracks prediction accuracy over rolling windows of 100 incidents, enabling performance monitoring and fine-tuning of the diagnostic triage process.

3.3.4 Component Integration

Integration with predictive maintenance is achieved through a REST API that queries the vehicle's health status in real time, retrieving key parameters such as battery voltage, coolant temperature, active fault codes, and predictive alerts. These data points are then fed directly into the diagnostic triage engine, allowing the system to enhance its service-type probability estimates by combining immediate incident reports with underlying vehicle health trends, thereby improving dispatch accuracy and reducing the likelihood of mismatches.

Integration with Geo-Intelligence enhances the dispatch framework by retrieving traffic impact scores on a 1–10 scale for each incident location, allowing the system to quantify congestion effects in real time. These scores are then incorporated as externality weights within the Expected-Cost Minimization (ECM) cost function, ensuring that provider assignments not only minimize mismatch risks but also account for broader traffic implications, thereby improving both efficiency and urban mobility outcomes.

Integration with the Compliance Component ensures that the system adheres to privacy and accountability standards by anonymizing all incident data before storage, requiring explicit user consent for accessing OBD-II telemetry, and maintaining comprehensive audit logs for every dispatch decision. This approach safeguards sensitive information, reinforces user trust, and provides a transparent record of system actions, thereby aligning the intelligent dispatch framework with ethical and regulatory expectations.

3.4 Tools and Technologies

Category	Technology	Purpose	Justification
Backend	Node.js + Express.js + TypeScript	API server and business logic	High performance async I/O for real-time dispatch
Database	PostgreSQL	Primary data store	ACID compliance, geospatial extensions
Cache	Redis	Real-time state management	Low-latency provider availability tracking
Message Queue	RabbitMQ	Async event processing	Reliable dispatch notification delivery
Maps API	Google Maps Platform	ETA calculation, routing	Industry-standard accuracy, Sri Lanka coverage
ML Framework	Python + scikit-learn	Bayesian learning, analytics	Mature statistical modeling ecosystem
Simulation	Python + SimPy	Dispatch simulation	Discrete event simulation for evaluation
Testing	Jest + Supertest	Unit and integration testing	Comprehensive Node.js testing framework
Containerization	Docker + Docker Compose	Deployment packaging	Consistent deployment across environments

Table 4 - Tools and Technologies

3.5 Data Collection

Incident type distributions used in the synthetic dataset were validated through informal consultations with local roadside assistance providers and insurance representatives, confirming that the proportions align with observed patterns in the Sri Lankan context. However, since no centralized digital incident database exists in Sri Lanka, the full dataset was synthetically generated using these validated distributions to ensure the availability of all fields required for simulation-based evaluation.

3.6 Simulation and Evaluation Plan

The simulation design employs a discrete event simulation framework built in Python using SimPy, covering the Colombo Metropolitan Area, the Southern Expressway corridor, and selected A-roads to capture diverse traffic and incident conditions. The model spans 14 simulated days, with peak and off-peak hours modeled separately to reflect realistic temporal variations in roadside emergencies and generates 1,000 incidents following empirically grounded distributions.

The fleet consists of 50 service units, including 8 heavy tow trucks, 15 light tow trucks, 18 mobile mechanics, 5 fuel delivery vehicles, and 4 locksmiths, representing the typical mix of roadside assistance resources available in urban and highway contexts. To incorporate reliability and performance heterogeneity, provider trust scores are modeled using a normal distribution with a mean of 0.75 and a standard deviation of 0.15, ensuring that historical performance influences dispatch decisions.

Metric	Definition	Target
Mismatch Rate	% of dispatches where provider could not resolve the incident	≤15% (vs. 25–35% baseline)
Mean Resolution Time	Average time from request to resolution	≤25 minutes
First-Time Resolution Rate	% of incidents resolved without re-dispatch	≥85%
Mean Wasted Provider Time	Average time providers spend on mismatched dispatches	≤10 minutes
Traffic Externality Cost	Total impact_score × resolution_time across all incidents	Minimize
Diagnostic Accuracy	% where highest-probability diagnosis matched actual	≥70%
Triage Entropy Reduction	Reduction in uncertainty from user-reported to triage output	≥40% reduction
OBD Enhancement	Accuracy improvement when OBD data available vs. questionnaire only	≥10% improvement
Learning Convergence	Number of incidents to reach stable accuracy	≤200 incidents

Table 5 - Evaluation Metrics and Targets

3.7 Ethical Considerations

This research uses synthetically generated incident data for simulation evaluation. No real personal data is collected or processed during the research phase. The platform design incorporates PDPA compliance principles including purpose limitation, data minimization, and explicit user consent mechanisms. For any future pilot testing with real users, informed consent will be obtained, and all data will be anonymized.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The Kaduna.lk platform follows a modular microservices architecture where four independent components interact through well-defined REST API interfaces. The Dispatch Optimization service is the core decision-making engine that coordinates with the other three components.

Dimension	Dispatch Optimization
Core Question	Which provider should respond to this incident?
Research Domain	Operations Research, Vehicle Routing Problem
Algorithm/ Method	ECM, Bayesian Learning, Decision Trees
Output	Provider assignment + ETA
Evaluation	Mismatch rate, resolution time

Table 6 - Component Boundary

Dispatch Service Internal Architecture

The Dispatch Service is internally organized into four components: Incident Manager that orchestrates the full lifecycle from request to resolution. Diagnostic Triage Engine that processes questionnaire responses and OBD-II data; Dispatch Optimizer that runs the ECM algorithm and Bayesian Learning Engine that updates diagnostic priors from feedback.

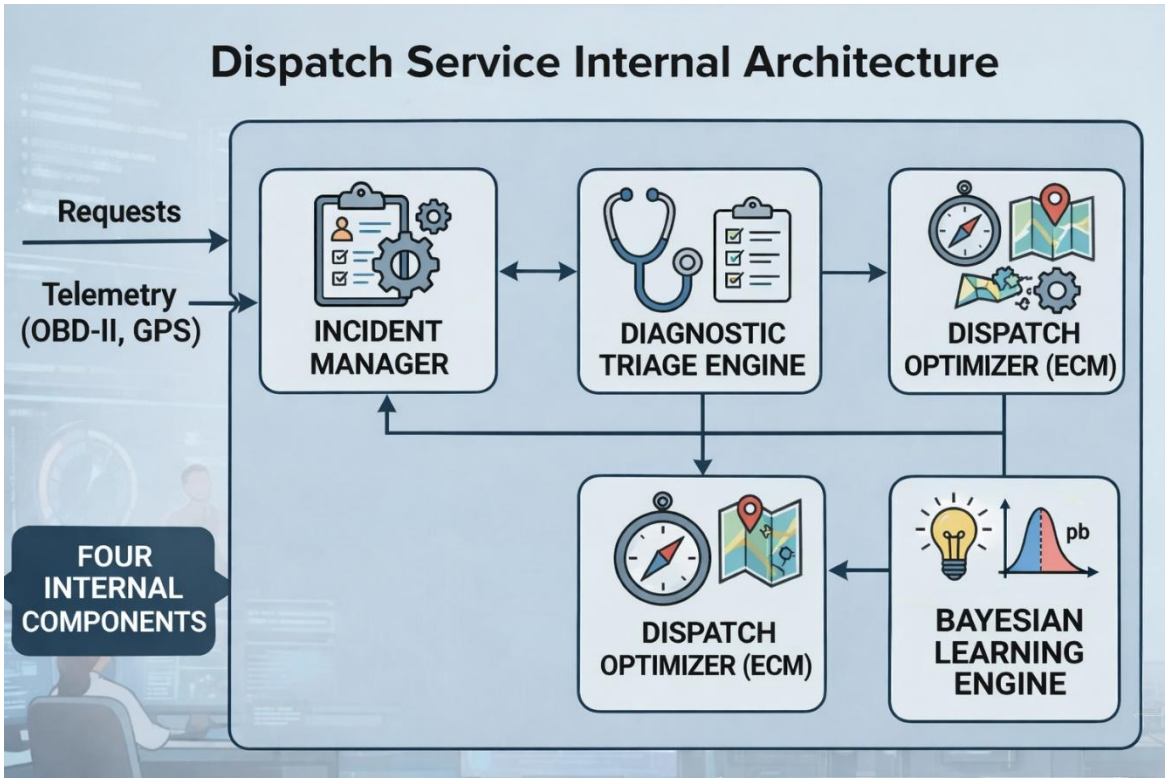


Figure 1- Dispatch Service Internal Architecture

Data Flow

The Dispatch Optimization system begins when a driver submits an incident request with GPS location, after which the Diagnostic Triage Engine combines structured questions with OBD-II data to generate a probability distribution of possible service needs. The Geo-Intelligence module then provides traffic impact scores, which feed into the Expected-Cost Minimization (ECM) optimizer. This optimizer evaluates providers by considering expected costs, mismatch penalties, traffic externalities, and trust scores, selecting the optimal provider in under 500ms. The system dispatches the request, monitors provider acceptance and progress, and keeps the driver updated. Once resolved, feedback on service type, resolution time, mismatch status, and user rating is collected, and Bayesian priors are updated with a decaying learning rate. Together, the four stages—Diagnostic Triage, Stochastic Dispatch, Monitoring, and Adaptive Feedback—form a continuous cycle where each incident improves future decisions, ensuring efficient, accurate, and adaptive roadside assistance.

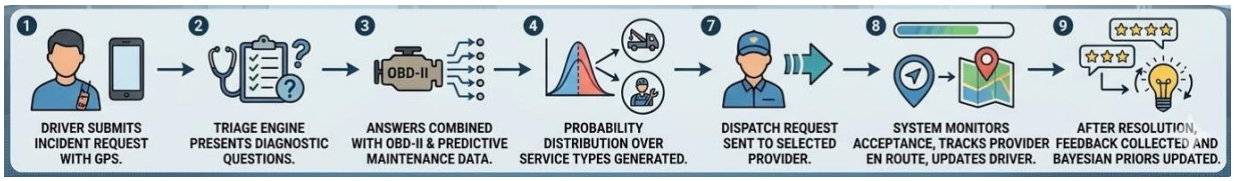


Figure 2 - Dispatch Service Data Flow

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Stakeholder	Primary Need
Stranded Drivers	Request emergency assistance quickly, receive accurate ETA, get the right help on first dispatch
Service Providers (Tow trucks, Mechanics)	Receive clear dispatch requests with incident context, minimize wasted trips
Traffic Authorities	Reduce incident-induced congestion through faster clearance
Platform Administrators	Monitor system performance, configure dispatch parameters

Table 7 - Stakeholder Identification

5.2 Functional Requirements

ID	Requirement	Priority
FR-01	Allow drivers to submit roadside assistance requests with GPS location and vehicle information	High
FR-02	Present a structured diagnostic questionnaire (5–8 questions) to gather diagnostic information	High
FR-03	Retrieve real-time OBD-II data from the Predictive Maintenance component for connected vehicles	High
FR-04	Generate probability distributions over possible service types based on triage and OBD data	High
FR-05	Retrieve traffic impact scores from the Geo-Intelligence component for each incident location	High
FR-06	Execute ECM dispatch optimization to select the optimal provider under uncertainty	High
FR-07	Transmit dispatch requests to selected providers with incident details and expected service type	High
FR-08	Track provider acceptance/rejection within configurable timeout (default: 120 seconds)	High
FR-09	Automatically re-dispatch to next-ranked provider if selected provider declines or times out	High
FR-10	Monitor provider location and update ETA in real-time using Maps API	Medium
FR-11	Allow providers to report on-scene findings and trigger escalation if needed	High
FR-12	Record incident resolution details including actual service type and resolution time	High
FR-13	Update diagnostic probability priors based on resolution outcomes using Bayesian learning	Medium
FR-14	Send real-time status notifications to drivers throughout the incident lifecycle	High

Table 8 - Functional Requirements

5.3 Non-Functional Requirements

ID	Category	Requirement	Target
NFR-01	Performance	Dispatch decision latency	< 500ms from triage to provider selection
NFR-02	Performance	Triage question response time	< 200ms per question transition
NFR-03	Performance	Concurrent incident handling	Support 100+ simultaneous incidents
NFR-04	Availability	System uptime	99.5% availability
NFR-05	Availability	Graceful degradation	Function with reduced features if OBD/Geo-Intel unavailable
NFR-06	Security	Authentication	JWT-based authentication for all API endpoints
NFR-07	Security	Data encryption	TLS 1.3 in transit, AES-256 at rest
NFR-08	Privacy	PDPA compliance	Full compliance with Sri Lanka PDPA
NFR-09	Privacy	Data minimization	Collect only data necessary for dispatch
NFR-10	Usability	Triage completion time	< 60 seconds for full questionnaire
NFR-11	Maintainability	Code coverage	Minimum 80% unit test coverage

Table 9 - Non-Functional Requirements

6. COMMERCIALIZATION PLAN

- **Target Market:** Daily commuters in urban Sri Lanka (Colombo, Kandy, Galle metropolitan areas), commercial fleet operators, insurance companies offering roadside assistance, and traffic management authorities.

- **Value Proposition:** The first uncertainty-aware dispatch platform for Sri Lankan roads. Unlike existing manual systems, Kaduna.lk's dispatch engine does not just find the nearest provider—it figures out what service is needed, picks the right provider for the job, and prioritizes incidents that will cause the most traffic disruption. This means fewer wasted trips, faster resolution, and less congestion.

- **Revenue Model:** Freemium model for individual drivers (free basic requests, premium for priority dispatch and real-time tracking). B2B subscription for fleet operators and insurance companies (API access to dispatch optimization and analytics). Government licensing for traffic authority integration.

- **Cost Estimation:** Development costs are primarily human effort (4 team members × 10 months, academic project). Infrastructure costs include cloud hosting, Google Maps API usage, and domain registration. Total estimated development cost: LKR 120,000.

- **Pricing Strategy:** Free tier for basic users; LKR 500/month for premium individual users; LKR 25,000–50,000/month for fleet operators (based on fleet size); custom pricing for government and institutional clients.

- **Competitive Advantage:** First-mover advantage in Sri Lankan market; proprietary stochastic dispatch algorithm with diagnostic triage; validated through simulation; integrated platform approach combining dispatch, traffic intelligence, predictive maintenance, and compliance.

- **IP Considerations:** The UADO framework, ECM algorithm, and diagnostic triage methodology may be eligible for intellectual property protection. Research findings will be published as academic contributions. The software codebase will be maintained as a private repository.

7. BUDGET AND JUSTIFICATION

Category	Item	Estimated Cost (LKR)	Justification
Cloud Services	AWS/GCP Development Environment (12 months)	30,000	Backend hosting, database, testing
Cloud Services	Redis Cloud (12 months)	6,000	Real-time state management for dispatch
Cloud Services	MongoDB Atlas (optional, 12 months)	9,000	Document storage for analytics
API Services	Google Maps Platform (12 months)	30,000	ETA calculation, routing, geocoding
API Services	Weather API (12 months)	5,000	Weather context for incidents (enhancement)
Dev Tools	GitHub Pro (if needed, 12 months)	10,000	Private repos, extended CI/CD minutes
Dev Tools	Postman Team (6 months)	5,000	API development and testing
Hardware	Personal laptop (existing)	0	Already owned
Hardware	OBD-II test device	0	Borrowed from Predictive Maintenance component
Documentation	Thesis printing and binding (5 copies)	7,500	Final submission
Documentation	Poster/presentation materials	4,500	Final presentation
Contingency	Unexpected expenses (10%)	13,000	Buffer for unforeseen costs
	TOTAL	120,000	

Table 10 - Budget Breakdown

8. WORK BREAKDOWN STRUCTURE (WBS)

WP	Task	Responsible	Timeline	Deliverables
WP1	Literature Review & Research Design	Janukshan	Month 1–3	Literature survey, system design, API specifications
WP2	Diagnostic Triage Engine	Janukshan	Month 4–5	Decision tree, OBD integration, probability estimation
WP3	ECM Dispatch Optimizer	Janukshan	Month 5–7	Mathematical formulation, ECM implementation, baselines
WP4	Bayesian Learning Engine	Janukshan	Month 7–8	Feedback collection, prior updates, convergence monitoring
WP5	Component Integration	All members	Month 8–9	API integration with Geo-Intel and Predictive Maintenance
WP6	Simulation & Evaluation	Janukshan	Month 9–11	Dataset generation, simulation runs, statistical analysis
WP7	Documentation & Thesis	Janukshan	Month 11–12	Thesis writing, presentation preparation, final submission

Table 11 – Work Breakdown

9. GANTT CHART

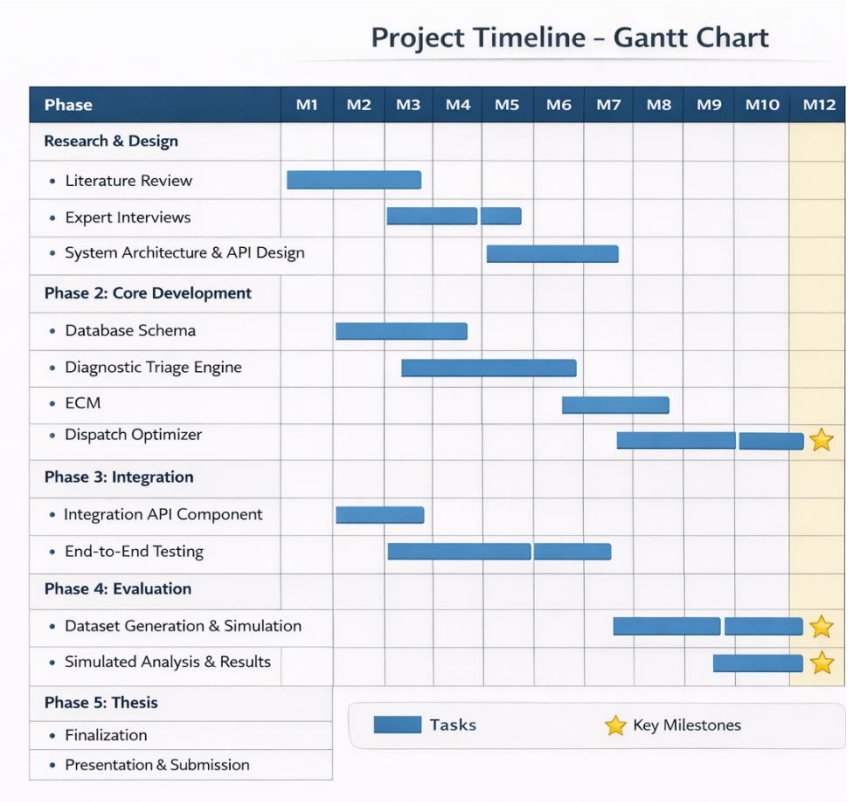


Figure 3 - Gantt Chart

Key Milestones:

Month 3: Design Completed, Architecture approved, APIs specified

Month 7: Core Developments completed, all algorithms implemented and unit tests.

Month 9: Integration completion with all components and end to end tests passing.

Month 11: Evaluation completion with simulation results documented.

Month 12 : Final submission with thesis and presentation ready.

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11. APPENDICES

Appendix A: UADO Framework Overview

The Uncertainty-Aware Dispatch Optimization framework operates as a pipeline that transforms an incident report into an optimal dispatch decision through three stages:

- **Stage 1 – Diagnostic Triage:** Gather information from the driver and vehicle, then produce a probability distribution over possible service types.
- **Stage 2 – Stochastic Dispatch Optimization:** Select the provider that minimizes expected total cost, accounting for all possible service outcomes, mismatch penalties, and traffic externality costs.
- **Stage 3 – Adaptive Feedback:** After resolution, update diagnostic models based on what service was needed, so the system improves over time.

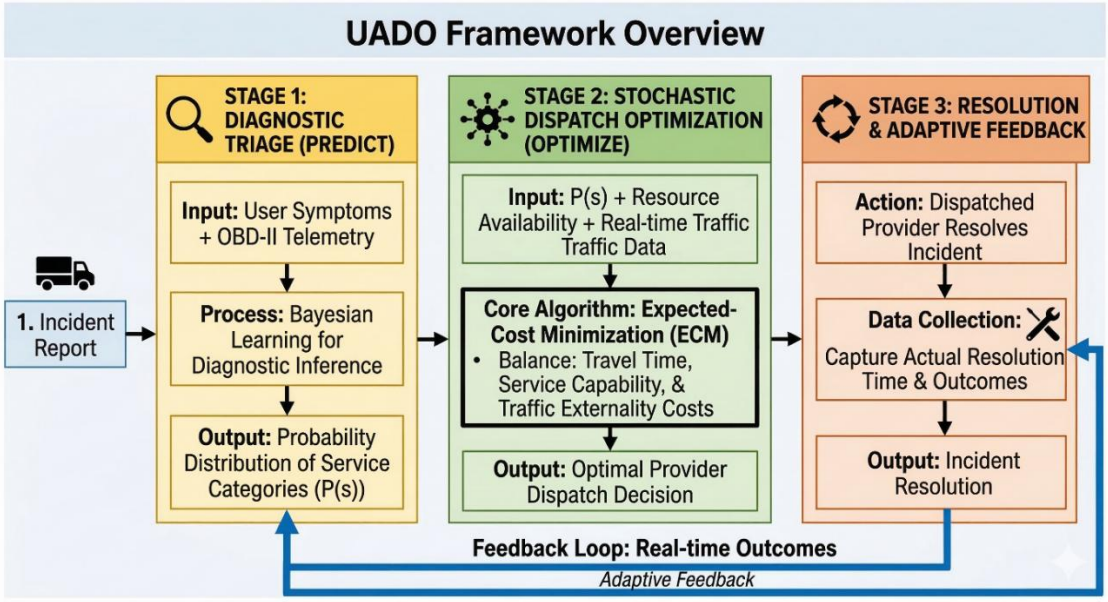


Figure 4 - UADO Framework Overview

Appendix B: Diagnostic Triage and Worked Example

A.1 Diagnostic Triage Structure

The diagnostic triage system addresses a fundamental question: what service does this incident need? Rather than accepting the user's self-reported category at face value, the system gathers structured information to produce a probability distribution over possible service types. The key insight is that users often do not know what is wrong with their vehicle. A driver who reports "my car won't start" could have a dead battery (45% of such cases), a starter motor failure (25%), a fuel pump failure (15%), an alternator failure (10%), or something else (5%). By asking target diagnostic questions and integrating OBD-II data, the system can significantly narrow this distribution.

Question Number	Question	Options	Diagnostic Purpose
Q1	Can you start the engine?	Yes / No / Partially (cranks but won't start)	Primary branch: running vs. not running
Q2	Do you hear any sounds when trying to start?	Clicking / Grinding / Nothing / Normal cranking	Distinguishes battery vs. starter vs. fuel/ignition
Q3	Are any dashboard warning lights on?	Battery / Engine / Oil / Temperature / Multiple / None	Identifies specific system failures
Q4	Is there visible damage to the vehicle?	Yes – Minor / Yes – Major / No	Determines if towing is likely needed.
Q5	Can the vehicle move at all?	Yes, normally / Yes, with difficulty / No / Unsafe to try	Determines if towing is required
Q6	Have you noticed any unusual smells?	Burning / Fuel / Smoke / None	Identifies fire risk, fuel leaks
Q7	What is your current location type?	Highway / Main road / Residential / Rural / Parking	Affects urgency and provider access
Q8	Is anyone injured or is the situation unsafe	Yes / No / Unsure	Determines incident urgency level for dispatch prioritization

Table 12 - Diagnostic Triage Sample Question Set

A.2 Worked Example

Scenario: A driver on Galle Road (A2) reports "my car won't start" at 8:45 AM on a weekday.

Questionnaire Responses:

- Q1: Can you start the engine? → **No**
- Q2: Sounds when trying to start? → **Clicking**
- Q3: Dashboard warning lights? → **Battery**
- Q4: Visible damage? → **No**
- Q5: Can the vehicle move? → **No**
- Q6: Unusual smells? → **None**
- Q7: Location type? → **Main road**
- Q8: Anyone injured? → **No**

OBD-II Data Retrieved:

- Battery voltage: 11.2V (below normal 12.4V threshold)
- Active fault code: P0562 (System Voltage Low)
- No crash sensor activation

Diagnostic Engine Processing:

1. Q1=No + Q2=Clicking → strong battery indicator (base P(BATTERY) = 0.60)
2. Q3=Battery light → reinforces battery diagnosis (+0.10)
3. OBD: P0562 + low voltage → high confidence battery/charging issue (+0.15)
4. No visible damage → standard dispatch priority"

Output Distribution:

- P(BATTERY_JUMP) = 0.45
- P(STARTER_MOTOR) = 0.20
- P(BATTERY_REPLACE) = 0.15
- P(FUEL_PUMP) = 0.12
- P(OTHER) = 0.08
- Confidence: 0.72
- Recommended: MECHANIC_MOBILE

Appendix B: Stochastic Dispatch Optimization

An incident occurs at a specific location, where diagnostic triage assigns probabilities to different required service types. Available providers—such as tow trucks or mechanics—are characterized by their location, service type, capabilities, and trust score. The challenge is to select the most suitable provider to dispatch, optimizing the decision under uncertainty while ensuring efficiency and reliability

Tow trucks can resolve more types of issues (battery, flat tire, towing), while mechanics are limited. This asymmetry is key when diagnosis is uncertain.

Cost definition:

- $Cost(p, i) = \Sigma_k P(type_k \mid triage) \times Resolution_Cost(p, i, type_k) + \lambda \times impact_i \times E[Resolution_Time(p, i)]$

Resolution Cost:

- If provider can handle the issue → travel time + service time.
- If provider cannot handle → wasted trip + assessment delay + re-dispatch + correct provider's travel + service time.

Traffic impact term: Incidents on busy roads are weighted more heavily, prioritizing faster resolution to reduce externalities.

Provider Type	Battery	Flat Tire	Fuel	Lockout	Mechanic Fix	Tow Light	Tow Heavy
Mobile Mechanic	✓	✓	✗	✗	✓	✗	✗
Fuel Delivery	✗	✗	✓	✗	✗	✗	✗
Locksmith	✗	✗	✗	✓	✗	✗	✗
Light Tow Truck	✓	✓	✗	✗	✗	✓	✗
Heavy Tow Truck	✓	✓	✗	✗	✗	✓	✓

Table 13 - Provider Type Capability Matrix

Appendix C: ECM Algorithm Pseudocode

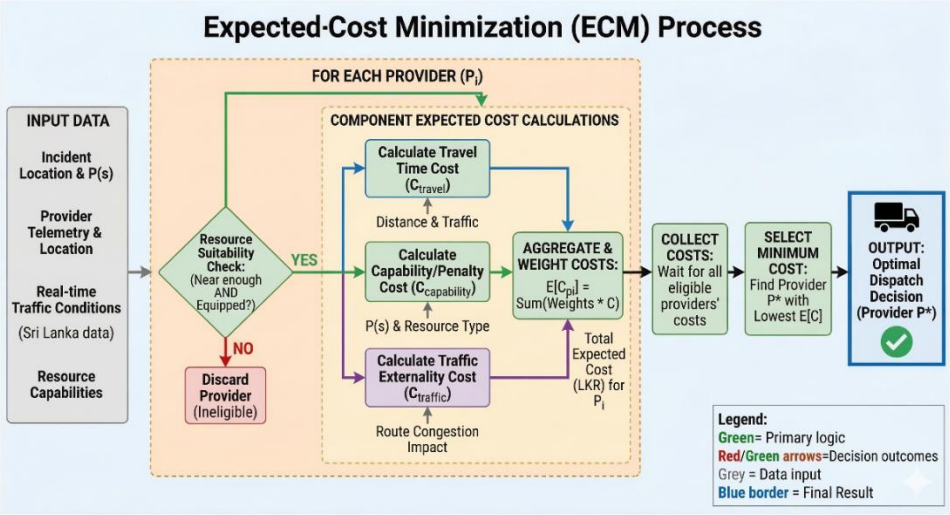


Figure 5 - Expected Cost Minimization Process

Complexity Analysis

The ECM algorithm runs in O(n × k) time where:

- n = number of available providers (typically 5–50 in a district)
- k = number of service types with non-negligible probability (typically 3–5)

For a typical scenario with 30 available providers and 5 probable service types, the algorithm evaluates 150 cost calculations, each involving a Maps API distance lookup (cached) and arithmetic operations. This ensures sub-500ms dispatch decisions even under load.

Appendix D: Bayesian Learning Algorithm Specification

C.1 Feedback Data Structure

Every resolved incident generates a feedback record with the following structure:

```
{
  "incident_id": "INC-2026-03-12-001",
  "triage_input": {
    "questionnaire_responses": {
      "Q1": "NO",
      "Q2": "CLICKING",
      "Q3": "BATTERY",
      "Q4": "NO",
      "Q5": "NO",
      "Q6": "NONE",
      "Q7": "MAIN_ROAD",
      "Q8": "NO"
    },
    "obd_data": {
      "battery_voltage": 11.2,
      "fault_codes": ["P0562"]
    }
  },
  "predicted_distribution": {
  "BATTERY_JUMP": 0.45,
  "STARTER_MOTOR": 0.20,
  "BATTERY_REPLACE": 0.15,
  "FUEL_PUMP": 0.12,
  "OTHER": 0.08
  },
  "predicted_service_type": "BATTERY_JUMP",
  "predicted_confidence": 0.72,
  "actual_service_type": "BATTERY_JUMP",
  "was_match": true,
  "provider_dispatched": "P_007",
  "provider_type": "MECHANIC_MOBILE",
  "resolution_time_minutes": 18,
  "re_dispatches": 0,
  "user_rating": 4.5,
  "timestamp": "2026-03-12T09:15:00Z"
  }
```

C.2 Bayesian Updating Formula

```
P(diagnosis | symptoms)_new = P(diagnosis | symptoms)_old × likelihood_adjustment

Where:
  likelihood_adjustment =
    IF actual_diagnosis == predicted_diagnosis:
      1 + learning_rate × (1 - confidence)
    ELSE:
      1 - learning_rate × confidence
```

Appendix E: UI Mockups

